The exact six-dimensional bad-science matrix optimum

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Abstract

For a real 6×6 matrix A whose rows have Euclidean norm one, let $\beta(A)$ be the average of $\|Ax\|_\infty$ over the sign cube. We prove that the exact maximum is $1 + \sqrt{3}/2$, thereby proving optimality of Steinerberger's six-dimensional construction. The upper bound follows from an affine bound for the length of the sum of an antipodal-free set of cube vertices. Coordinate compression reduces the small-cardinality part of this bound to 7,867 order ideals, checked by a short exhaustive integer computation; cube orthogonality handles all remaining cardinalities. The enumeration, its completeness proof, and the exact arithmetic checks are included.

1 The problem and the result

Put $Q = \{-1, 1\}^6$. For $A \in \mathbb{R}^{6\times 6}$ with rows $a_1, \dots, a_6$ satisfying $\|a_i\|_2 = 1$, define

$$\beta(A) = \frac{1}{64} \sum_{x\in Q} \|Ax\|_\infty , \qquad \beta_6 = \max_{\|a_i\|_2=1} \beta(A).$$

The maximum exists because the product of six unit spheres is compact and the objective is continuous.

Theorem 1. *The exact six-dimensional optimum is*

$$\boxed{\beta_6 = 1 + \frac{\sqrt{3}}{2}}.$$

An attaining matrix is

$$A_* = \frac{1}{2\sqrt{3}} \begin{pmatrix} -\sqrt{3} & 0 & \sqrt{3} & -\sqrt{3} & 0 & -\sqrt{3} \\ \sqrt{3} & 0 & -\sqrt{3} & \sqrt{3} & 0 & -\sqrt{3} \\ \sqrt{3} & 0 & \sqrt{3} & \sqrt{3} & 0 & \sqrt{3} \\ 2 & -2 & 0 & -2 & 0 & 0 \\ -\sqrt{3} & 0 & -\sqrt{3} & -\sqrt{3} & 0 & \sqrt{3} \\ 2 & 2 & 0 & -2 & 0 & 0 \end{pmatrix}. \tag{1}$$

Steinerberger introduced the problem and gave precisely the matrix (1), together with its objective value [1, Section 2.6]. Albors et al. proved exact optimality results in dimensions at most four and developed a structural theory [2]. Sinha studied further structural and asymptotic aspects [3]. A public five-dimensional proof uses an affine bound for sums of antipodal-free sets and sums it over oriented winning-row cells [4]. That argument motivates the approach here. The new step is the six-dimensional affine bound in Lemma 6, which gives a matching upper bound for the known construction. The proof is computer-assisted only in the small finite enumeration of Lemma 5.

2 Oriented cells and cube sums

A subset $S \subseteq Q$ is *antipodal-free* if it does not contain both x and $-x$. Write

$$v(S) = \sum_{x \in S} x.$$

An *oriented cell family* consists of six sets $S_1, \dots, S_6$ such that each antipodal pair $\{x, -x\}$ contributes exactly one vertex to exactly one cell. Empty cells are allowed. In particular, every cell is antipodal-free and $\sum_i |S_i| = 32$.

Proposition 2. *The matrix optimization has the finite reformulation*

$$32\beta_6 = \max_{(S_1,\dots,S_6)} \sum_{i=1}^{6} \|v(S_i)\|_2 ,$$

where the maximum is over all oriented cell families.

Proof. Given A, choose one representative of each antipodal pair. Assign it to a row attaining $\max_i |a_i \cdot x|$, breaking ties by the least row index. Choose its sign so that its inner product with the assigned row is nonnegative; either sign is allowed when that product is zero. The resulting family satisfies

$$32\beta(A) = \sum_{i=1}^{6} a_i \cdot v(S_i) \le \sum_{i=1}^{6} \|v(S_i)\|_2 .$$

Here the factor 32 follows from $\|A(-x)\|_\infty = \|Ax\|_\infty$. Conversely, given a family, choose $a_i = v(S_i)/\|v(S_i)\|_2$ for each nonzero sum, and any unit vector when the sum is zero. For the resulting matrix,

$$32\beta(A) = \sum_i \sum_{x \in S_i} \|Ax\|_\infty \ge \sum_i \sum_{x \in S_i} a_i \cdot x = \sum_i \|v(S_i)\|_2 .$$

Taking maxima proves both inequalities, without any assumption about uniqueness of a maximizing row. □

Lemma 3 (Orthogonality bound)**.** *If $S \subseteq Q$ is antipodal-free and $m = |S|$, then*

$$\|v(S)\|_2^2 \le 32m.$$

Proof. The claim is immediate if $m = 0$ or $v = v(S) = 0$. Otherwise cube orthogonality gives

$$\sum_{x \in Q} (v \cdot x)^2 = 64 \|v\|_2^2 .$$

Since S and $-S$ are disjoint, Cauchy–Schwarz gives

$$64 \|v\|_2^2 \ge 2 \sum_{x \in S} (v \cdot x)^2 \ge \frac{2}{m} \left(\sum_{x \in S} v \cdot x \right)^2 = \frac{2}{m} \|v\|_2^4 .$$

Division by $2\|v\|_2^2/m$ proves the assertion. □

3 Small sets by coordinate compression

Encode a cube vertex by the integer $b \in \{0, \dots, 63\}$ whose jth binary digit is $(1 - x_j)/2$, for $j = 0, \dots, 5$ after relabeling coordinates. A collection of codes is an *order ideal* if clearing any one of the set bits of a member produces another member.

Lemma 4 (Compression)**.** *For every $S \subseteq Q$ there is an order ideal I of the same cardinality such that $\|v(S)\|_2 \le \|v(I)\|_2$.*

Proof. First reflect coordinates so that every coordinate of $v(S)$ is nonnegative. These reflections preserve its norm and cardinality. If a vertex with coordinate -1 is present and its partner obtained by changing that coordinate to $+1$ is absent, replace the former by the latter. This increases one coordinate of the sum by 2 and leaves all other coordinates unchanged. The coordinates of the sum stay nonnegative, so its squared norm does not decrease.

Continue such replacements in any order. The total number of negative coordinates among all vertices decreases by one at every step, so the process terminates. At termination every required predecessor is present, which is exactly the order-ideal property. The replacements preserve cardinality. The terminal set is allowed to contain antipodal pairs: this lemma bounds arbitrary subsets, and does not require antipodal-freeness. □

Lemma 5 (Exact small-set table)**.** *For $0 \le m \le 11$, the maximum of $\|v(S)\|_2^2$ over all m-element subsets of Q is the number M_m in the following table. The row c_m gives the number of order ideals of that cardinality.*

m	0	1	2	3	4	5	6	7	8	9	10	11
M_m	0	6	20	38	64	86	116	150	192	214	244	278
c_m	1	1	6	15	35	75	156	306	605	1110	2045	3512

Proof. By Lemma 4, it suffices to maximize over order ideals. The following complete Python code performs that finite enumeration, using integers throughout. A set of codes is stored as a 64-bit mask; $P[b]$ has a bit at every immediate predecessor of b. The argument t is the largest code already inserted.

```
P = [sum(1 << (b ^ (1 << j)) for j in range(6)
         if b & (1 << j)) for b in range(64)]
X = [tuple(1 - 2*((b >> j) & 1) for j in range(6))
     for b in range(64)]
M, C = [-1]*12, [0]*12
def visit(mask, t, m, v):
    C[m] += 1
    M[m] = max(M[m], sum(z*z for z in v))
    if m == 11:
        return
    for b in range(t+1, 64):
        if (P[b] & mask) == P[b]:
            w = tuple(v[j] + X[b][j] for j in range(6))
            visit(mask | (1 << b), b, m+1, w)
visit(0, -1, 0, (0,)*6)
assert M == [0,6,20,38,64,86,116,150,192,214,244,278]
assert C == [1,1,6,15,35,75,156,306,605,1110,2045,3512]
```

Every visited set is an ideal: the insertion test checks all immediate predecessors, and the preceding set already contains their predecessors. Conversely, list the elements of any ideal in increasing

integer order. Every immediate predecessor of a code is a smaller integer, so each successive insertion passes the test. This list gives exactly one recursive path. Thus every ideal of size at most eleven is visited once, and the computation covers $\sum_m c_m = 7{,}867$ sets. The coordinate sum is updated by integer addition, so the calculated objective is exactly its squared norm. Finally, the initial code segments $\{0, \dots, m-1\}$ attain all the displayed values; thus the upper bounds are also maxima over arbitrary subsets. □

4 The affine bound and global optimality

Lemma 6 (Affine cube-sum bound)**.** *Every antipodal-free $S \subseteq Q$, with $m = |S|$, satisfies*

$$\|v(S)\|_2 \le L(m) := (2\sqrt{3} - 2)m + 16 - 8\sqrt{3}. \tag{2}$$

Equality can occur only when $m = 4$ or $m = 8$.

Proof. The slope and intercept of L are positive, so $L(m) > 0$ for $m \ge 0$. For $0 \le m \le 11$, use Lemma 5. Writing $a = 16 - 2m$ and $b = 2m - 8$, the squared gap is

$$L(m)^2 - M_m = (a^2 + 3b^2 - M_m) + 2ab\sqrt{3} = u_m + w_m\sqrt{3}.$$

Its coefficients are

m	0	1	2	3	4	5	6	7	8	9	10	11
u_m	448	298	172	74	0	−38	−52	−38	0	90	204	346
w_m	−256	−168	−96	−40	0	24	32	24	0	−40	−96	−168

The rational inequalities $5/3 < \sqrt{3} < 7/4$ show directly that each gap is positive except at $m = 4, 8$, where it vanishes. For example, the gap at $m = 0$ is strictly greater than $448 - 256(7/4) = 0$; the gap at $m = 6$ exceeds $-52 + 32(5/3) = 4/3$.

For $m \ge 12$, Lemma 3 applies. Let $D(t) = L(t)^2 - 32t$ for real t. Direct differentiation gives

$$D(12) = 448 - 256\sqrt{3} > 0, \qquad D'(12) = 96(2 - \sqrt{3}) > 0, \qquad D''(t) = 2(2\sqrt{3} - 2)^2 > 0.$$

It follows that $D(m) > 0$ for every $m \ge 12$. Combining the two ranges and taking nonnegative square roots proves (2) and the equality restriction. □

Proof of Theorem 1. For any oriented cell family, let $m_i = |S_i|$. Since $\sum_i m_i = 32$, Lemma 6 gives

$$\begin{aligned}\sum_{i=1}^{6} \|v(S_i)\|_2 &\le (2\sqrt{3} - 2)\sum_{i=1}^{6} m_i + 6(16 - 8\sqrt{3}) \\ &= 32 + 16\sqrt{3}.\end{aligned}$$

Proposition 2 yields $\beta_6 \le 1 + \sqrt{3}/2$.

For the matching lower bound, every row of (1) has squared norm one: four rows have four entries of absolute value $1/2$, and two rows have three entries of absolute value $1/\sqrt{3}$. For $x \in Q$, put $s = x_1 + x_4$ and $d = x_1 - x_4$. Up to absolute value, the first, second, third and fifth row products comprise all four expressions $(s \pm x_3 \pm x_6)/2$. Their maximum absolute value is $(|s|+2)/2$. The other two row products are $(d \pm x_2)/\sqrt{3}$, whose maximum absolute value is $(|d| + 1)/\sqrt{3}$. Consequently

$$\|A_* x\|_\infty = \begin{cases} 2, & x_1 = x_4, \\ \sqrt{3}, & x_1 = -x_4. \end{cases}$$

Each case occurs for 32 vertices. Thus $\beta(A_*) = (32 \cdot 2 + 32\sqrt{3})/64 = 1 + \sqrt{3}/2$. □

Corollary 7. *For every maximizing matrix and every tie-breaking rule, four oriented winning-row cells have size* 4 *and two have size* 8*.*

Proof. Equality in the upper bound forces $\|v(S_i)\|_2 = L(m_i)$ for each cell. Hence $m_i \in \{4, 8\}$. Six such integers sum to 32 exactly when two are 8 and four are 4. □

This cardinality restriction does not classify all attaining matrices; Theorem 1 determines the optimal value and gives one attaining matrix.

5 Reproducibility

The essential finite computation is fully displayed in Lemma 5. The accompanying Python standard-library script `src/verify_theorem.py` additionally checks the affine gaps with rational arithmetic, the row norms, and all 64 matrix images. From the reproduction package, run

```
python src/verify_theorem.py
```

using Python 3 without optimization flags. It reports the exact small-set maxima, the 7,867 ideals counted, and the image squared-norm counts 32 at 3 and 32 at 4.

A separate discovery program uses a different recursion: an ideal in the six-cube is a pair of nested ideals in the five-cube. It enumerates all 7,828,354 six-cube ideals and returns the same small-cardinality maxima. The shorter verifier is independent of that program and uses no third-party numerical library. The mathematical proof and these exact computations constitute the verification supplied here.

References

[1] Stefan Steinerberger, *Bad Science Matrices*, arXiv:2402.03205v2, 2024. arxiv.org/abs/2402.03205.

[2] Alex Albors, Hisham Bhatti, Lukshya Ganjoo, Raymond Guo, Dmitriy Kunisky, Rohan Mukherjee, Alicia Stepin, and Tony Zeng, *On the Structure of Bad Science Matrices*, Involve **19** (2026), 443–460; arXiv:2408.00933v2. doi:10.2140/involve.2026.19.443.

[3] Shridhar Sinha, *The Structure of Extremal Bad Science Matrices*, arXiv:2509.10580v1, 2025. arxiv.org/abs/2509.10580.

[4] trureturing, *The exact five-dimensional bad science optimum*, public research dossier and linked proof source, accessed 4 November 2026. `https://www.trureturing.org/research/albors-et-al-2024-bad-science-matrix-five/`.